

Java Practice Questions: Loops and Functions

These exercises help students practice Java looping statements (for, while) and user-defined functions (methods). Students should write complete Java programs and test them with different inputs.

1. Palindrome Number (Using Function)

- Write a Java program that checks whether a number is a palindrome or not.
- Create a function named `isPalindrome(int num)`.
- The function should return true if the number is palindrome, otherwise false.
- Example: 121 → Palindrome, 123 → Not Palindrome.

2. Sum of Digits (Using Loop and Function)

- Write a Java program to find the sum of digits of a number.
- Create a function `sumOfDigits(int num)`.
- Use a loop to extract each digit and calculate the total sum.
- Example: Input: 1234 → Output: 10.

3. Reverse a Number (Using Function)

- Write a Java program to reverse a given number.
- Create a function `reverseNumber(int num)`.
- Use a loop to reverse the digits.
- Example: Input: 1234 → Output: 4321.

4. Factorial of a Number

- Write a Java program to find the factorial of a number.
- Create a function `factorial(int n)`.
- Use a loop to calculate factorial.
- Example: Input: 5 → Output: 120.

5. Count Digits in a Number

- Write a Java program to count the number of digits in a given number.

- Create a function `countDigits(int num)`.
- Use a loop to determine how many digits are present.
- Example: Input: 45892 → Output: 5.